Tayyabah Rehman

AI / ML Developer | Deep Learning | Computer Vision | Python

your.email@example.com | github.com/your-username

PROFILE

MPhil Artificial Intelligence student with hands-on experience building and deploying machine learning and deep learning solutions. Proficient in Python and modern ML frameworks (PyTorch, TensorFlow, Scikit-learn) with a strong track record across classification, regression, clustering, computer vision, and NLP tasks. Currently completing two concurrent internships and an MPhil thesis on Automatic License Plate Recognition (95.11% exact-match accuracy). Eager to contribute to Asapp Studio's AI-driven products as an Associate AI/ML Developer.

EXPERIENCE

Machine Learning Intern — Dawood Technextgen (Remote)   May 2026 – Present

- Collect, clean, and preprocess real-world datasets for supervised and unsupervised ML pipelines.
- Perform feature engineering, EDA, and build classification / regression models using Scikit-learn.
- Document model performance metrics and maintain reproducible workflows — fully self-managed remotely.

Computer Vision Intern — AYK Private Limited (Onsite)   2025 · 1 Month

- Deployed YOLOv8 for real-time object detection and integrated PaddleOCR for text extraction from detected regions.
- Built a complete Python inference pipeline tested on live traffic footage; presented results to the technical team.

PROJECTS

Automatic License Plate Recognition (ALPR) — MPhil Thesis (In Progress)

Python · PyTorch · YOLOv8 · TrOCR · PaddleOCR · EasyOCR · OpenALPR

- End-to-end AI pipeline: data collection → preprocessing → model training → OCR evaluation on 1,522 image pairs.
- Benchmarked TrOCR, PaddleOCR, and EasyOCR; TrOCR achieved 95.11% exact-match accuracy.
- Integrated OpenALPR Snapshot Cloud API for real-time recognition.

Image Classification with CNN — Custom Fine-Tuning

Python · PyTorch · TensorFlow · CNN

- Fine-tuned pre-trained CNNs for custom image classification tasks; applied dropout and regularisation to reduce overfitting.
- Evaluated performance with accuracy, precision, recall, and confusion matrices.

Sign Language Digit Classification (0–9) & Binary Class Classification

Python · PyTorch · CNN · Scikit-learn

- Built CNN models for sign language digit recognition (0–9) and binary classification tasks.
- Applied feature optimisation and preprocessing pipelines to maximise model accuracy.

Fire and Smoke Detection from Video

Python · TensorFlow · CNN · LSTM · 3D-CNN · OpenCV

- Preprocessed video frame data; trained and compared CNN-LSTM, 2D-CNN, and 3D-CNN architectures.
- Evaluated detection accuracy and inference speed; deployed with OpenCV for real-time monitoring.

RNN for Text Generation

Python · PyTorch · RNN / LSTM

- Designed and trained a recurrent neural network for sequential text generation.
- Tuned hyperparameters (learning rate, hidden size, sequence length) to improve output coherence.

K-Means Clustering, Logistic Regression & Linear Regression
Python · Scikit-learn · Pandas · Matplotlib

- Implemented K-Means clustering for unsupervised segmentation tasks with visualised cluster boundaries.
- Built logistic and linear regression pipelines with hyperparameter optimisation; evaluated with R^2, RMSE, and accuracy.

Data Analysis, Visualisation & Feature Engineering Pipeline
Python · Pandas · NumPy · Matplotlib · Seaborn · Scikit-learn

- Designed end-to-end preprocessing pipelines: missing value imputation, encoding, normalisation, and feature engineering.
- Produced EDA dashboards and visualisation reports to communicate data insights clearly.

Effect of Regularisation and Dropout on Model Performance
Python · PyTorch · TensorFlow

- Conducted comparative experiments on L1/L2 regularisation and dropout across neural network architectures.
- Documented training/validation curves and generalisation trade-offs for different hyperparameter settings.

Face Recognition-Based Attendance System
Python · OpenCV · Deep Learning Face Embeddings

- Built automated data collection, preprocessing, and real-time face matching pipeline with attendance logging.

Video Caption Generation & Image Captioning
Python · PyTorch · CNN-LSTM · ResNet-50 · BLEU Score

- Built vision-language pipelines using CNN-LSTM and Two-Stream architectures; evaluated with BLEU metrics.

EDUCATION

MPhil in Artificial Intelligence (Ongoing)   2024 – 2026
University of the Punjab

Courses: Machine Learning, Deep Learning, Computer Vision, NLP, Pattern Recognition

Bachelor in Computer Science — Data Science Specialization   Jun 2024
University of Gujrat, Hafiz Hayat Campus

TECHNICAL SKILLS

Languages: Python

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn

ML Algorithms: Linear Regression, Logistic Regression, K-Means Clustering, CNN, LSTM, RNN, 3D-CNN, Two-Stream

Model Techniques: Hyperparameter Optimisation, Regularisation (L1/L2), Dropout, Fine-Tuning, Feature Engineering

Computer Vision: OpenCV, YOLOv8, RealESRGAN, DarkIR, HAT, X-Restormer+++

NLP / OCR: NLTK, TrOCR, PaddleOCR, EasyOCR, OpenALPR

Data Science: Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Cleaning, Preprocessing Pipelines

Evaluation: Accuracy, RMSE, MAE, R^2, PSNR, SSIM, LPIPS, BLEU, Confusion Matrix

Tools: Jupyter Notebook, Google Colab, PyCharm, GitHub, Kaggle, Roboflow

Mathematics: Linear Algebra, Calculus, Statistics, Probability

GitHub: github.com/your-username | Available to start immediately